

EuRepoC CYBER INCIDENTS

Exploratory Data Analysis Report

European Repository of Cyber Incidents • Global, Receiver, Attribution & Dyadic Datasets

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OVERVIEW

4,296 TOTAL INCIDENTS	75.3% ATTRIBUTED	195 RECEIVER COUNTRIES	2000–24 TIME SPAN
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This analysis examines trends across 4,296 documented cyber conflict dyads from 2000 to 2024, identifying temporal shifts in bilateral activity, primary initiator and target states, and the global geographic flow of operations. It further details the tactical methods defining these exchanges and highlights the specific country-to-country pairings that characterize the modern cyber landscape.

◆ KEY INSIGHT

Roughly 95.2% of dyadic operations are cross-border, with initiators and receivers situated in distinct nations. This metric underscores the transnational essence of contemporary cyber engagement, where physical proximity offers zero protection against offensive operations.

EXPLORATORY DATA ANALYSIS

01 Incidents Over Time

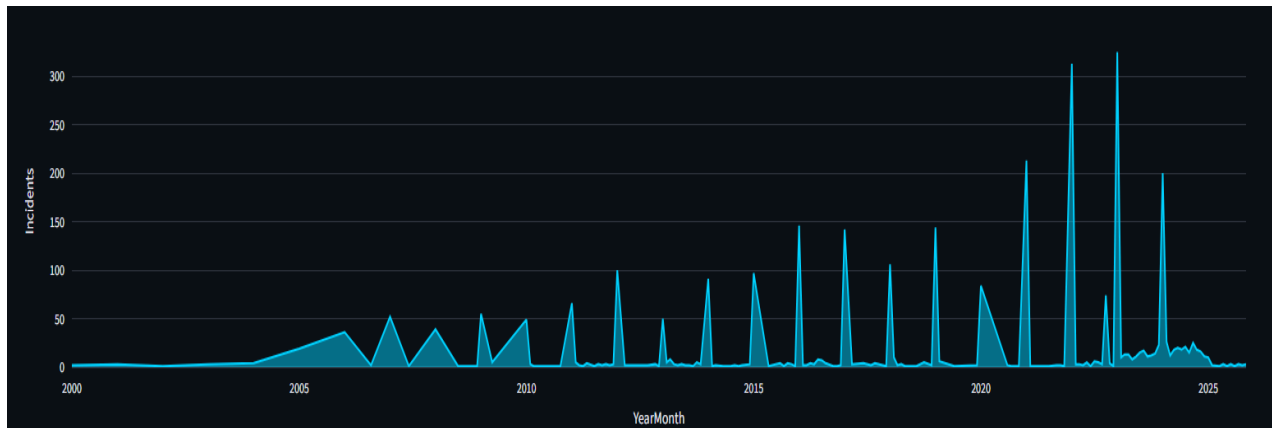


Figure 1. Number of recorded cyber incidents per year (2000–2024).

The incident timeline is punctuated by recurring, high-magnitude spikes that intensify both in frequency and amplitude after 2015, peaking aggressively during the 2021–2024 period.

- The regularity of these spikes suggests that cyber activity is highly event-driven—likely tied to specific geopolitical flashpoints or major international conflicts—rather than a continuous, steady-state threat environment.

◆ KEY INSIGHT

The incident timeline exhibits a clear 'ratchet effect,' where each successive spike establishes a higher baseline of operational activity rather than returning to previous levels of quiescence. This indicates a structural shift in the global cyber landscape: once new operational capacities or conflict vectors are established, they remain active, leading to a cumulative and permanent increase in daily threat volume.

02 Distribution of Operation Types

Distribution of Operation Types

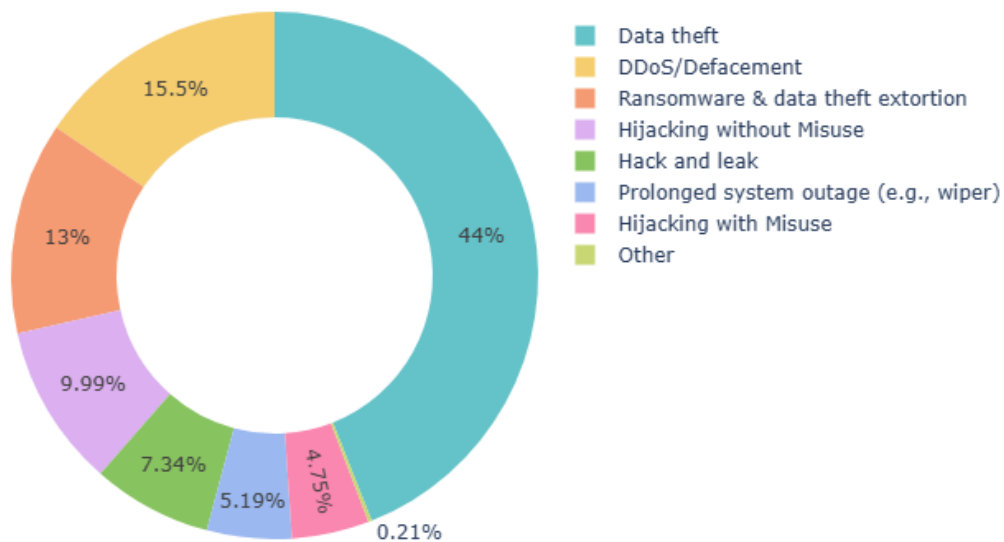


Figure 2. Operation types

The distribution of operation types is heavily skewed, with Data Theft alone accounting for more than half of all recorded incidents (52.8%).

◆ KEY INSIGHT

This extreme concentration suggests that while the cyber threat landscape is diverse, the majority of offensive activity is driven by a singular objective: the extraction of information rather than the disruption of services or infrastructure.

03 Top Receiver (Target) Countries

Dyadic Network Explorer

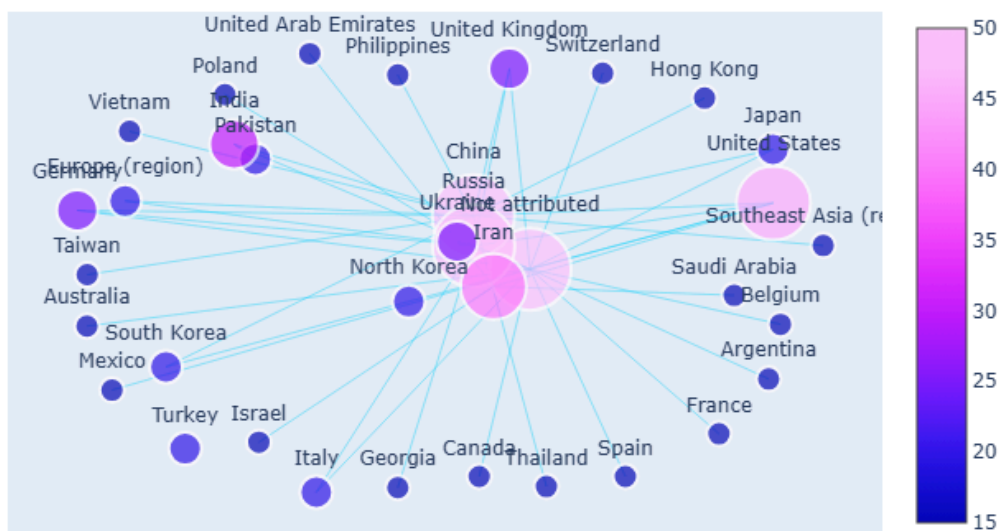


Figure 3. Initiator receiver graph connections

The network graph reveals a highly centralized hub-and-spoke topology, where a small cluster of major geopolitical powers accounts for a disproportionate volume of both global initiator and receiver activity. This concentrated structure confirms that global cyber conflict is not a diffuse, egalitarian phenomenon, but rather a contest dominated by a limited number of "super-nodes" that serve as the primary engines of systemic instability.

◆ KEY INSIGHT

The US-to-everyone-else gap is the single most striking asymmetry in the entire dataset, positioning the United States as the unique gravitational center of global cyber operations. While cyber conflict is ostensibly global, the network graph reveals that it is functionally anchored by a massive, disproportionate volume of activity directed at the U.S., making it the primary theater for international cyber competition.

04 Monthly Incidents by Initiator category

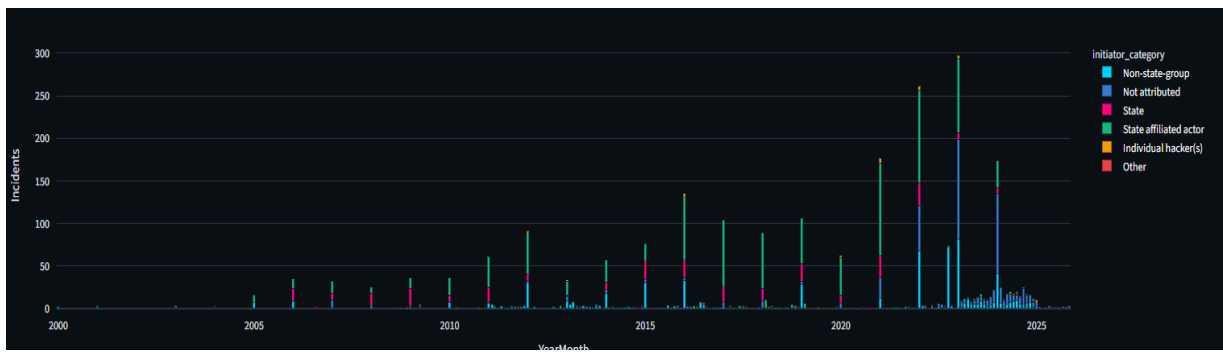


Figure 4. Distribution of incident types.

The surge in volume since 2021 is driven primarily by "State affiliated actors," creating a distinct "threat layer" that now dwarfs the baseline activity of non-state groups and individual hackers. This stratification confirms that the recent escalation in cyber incidents is not a random increase in activity, but a strategic expansion of state-led offensive operations, cementing their role as the primary force behind modern cyber instability.

KEY TAKEAWAYS

1. Incidents Over Time

- **Intensifying Spikes:** The timeline shows high-magnitude spikes that grew in frequency and amplitude after 2015, with aggressive peaking between 2021 and 2024.
- **Event-Driven Nature:** The regularity of these spikes suggests cyber activity is tied to specific geopolitical flashpoints or conflicts, rather than representing a steady-state threat.
- **The "Ratchet Effect":** Each successive spike establishes a higher baseline of operational activity, resulting in a cumulative, permanent increase in daily threat volume.

2. Distribution of Operation Types

- **Data Theft Dominance:** The landscape is heavily skewed, with Data Theft alone accounting for more than half (52.8%) of all recorded incidents.
- **Strategic Objective:** The extreme concentration on theft indicates that the majority of offensive activity is driven by information extraction rather than the disruption of services or infrastructure.

3. Dyadic Network Explorer

- **Centralized Topology:** The network operates as a hub-and-spoke model, dominated by a limited number of "super-nodes" that drive global systemic instability.
- **U.S. Asymmetry:** The United States acts as the unique gravitational center of global cyber operations, with a massive volume of activity directed at it compared to any other nation.

4. Monthly Incidents by Initiator Category

- **State-Affiliated Surge:** The post-2021 surge in volume is primarily driven by state-affiliated actors.
- **Stratified Threat Layer:** This growth creates a distinct "threat layer" that now dwarfs the baseline activity of individual hackers and non-state groups, marking a strategic expansion of state-led offensive operations.

5. Geospatial Threat Map

- **Geographic Disparity:** There is a clear asymmetry where attacks originate from a localized set of nations but target a wide range of global receivers.
- **Target of Choice:** Established global powers, particularly Western nations, are overwhelmingly positioned as the primary destination for these operations.

6. Intensity by Operation Type

- **Consequence vs. Frequency:** While Data Theft is the most frequent *method*, disruptive attacks that challenge state stability generate disproportionately higher *severity* levels.
- **Strategic Coercion:** The focus on "Critical infrastructure" and "State institutions" reveals that the primary theater of conflict is the foundational architecture of the state, used increasingly as a tool for strategic coercion.